

Learning Common Norms in Multi-Agent Systems

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1 INTRODUCTION

How can autonomous agents, each pursuing individual goals with limited information, come together to make collective decisions in complex environments? This question lies at the heart of multi-agent systems research and remains an open, critical challenge. In situations where agents must coordinate—whether by adopting shared strategies, interpreting observations collectively, or establishing mutual understanding—the formation of common knowledge, often embodied in shared norms, is essential. Without this foundation, agents risk misalignment, which can lead to inefficiencies or even system failure. Solving this problem is paramount, not just for advancing theory but for enabling high-stakes applications, from autonomous vehicle fleets to expansive, distributed AI networks.

The significance of this challenge is especially evident when compared to human societies. In social systems, common norms, cultural expectations, and institutional frameworks create shared understanding, enabling individuals to navigate complex interactions. Over time, these norms evolve to support coordinated action, even among people with diverse viewpoints and incomplete information. Just as common norms stabilize social behavior and support collaborative decision-making in society, common knowledge in multi-agent systems enables agents to work collectively toward shared goals, despite differences in individual observations or objectives.

Yet, unlike human societies—where norms and shared knowledge develop organically, adapting to the group’s diverse needs—current multi-agent systems lack established mechanisms for evolving shared norms. This gap presents a fundamental obstacle to scalable, effective coordination in multi-agent systems, especially in dynamic environments where agents must rapidly adapt to change.

This paper addresses the critical challenge of enabling agents in multi-agent systems to autonomously establish common norms. To achieve this, we explore reward structures as a mechanism for the learning and emergence of shared norms. Our objective is to design a reward structure that satisfies three essential properties:

- (1) Each agent selects actions that maximize its own reward within the specified reward structure.
- (2) Collectively, agents converge to a common norm.
- (3) The resulting common norm is (close to) socially optimal, maximizing (or nearly maximizing) the overall social welfare of the system.

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The reward structure we consider is inspired by social group dynamics and consists of three types of rewards that agents can choose to optimize:

- (1) *Personal Utility*: An agent's intrinsic rewards based on their preferences.
- (2) *Providing Utility to Others*: Agents extrinsic rewards gained from benefiting other agents. We refer to this reward as *reputation*.
- (3) *Providing Information*: Agents extrinsic rewards gained from sharing information about which norms lead to high reputation. We refer to this reward as *status*.

Each agent focuses on the reward type that provides the highest incentive. Depending on the reward they prioritize, agents perform distinct roles within the group. While different agents may take on different roles, we show that each reward type—and the corresponding role—is essential for a common norm to emerge. Notably, the presence of all three reward types is necessary to support the emergence of a shared norm.

Intuitively, if all agents optimize their reputation by choosing norms that provide the highest utility to others, a common norm that maximizes social welfare should emerge. To assess their reputation, agents interact with others to gauge how well their norms are received. However, reputation learning faces two main challenges: first, interacting agents may not provide accurate feedback due to limitations in knowledge or inherent biases. Second, agents would need to interact with all other agents to obtain an accurate assessment of their reputation, which becomes infeasible in large systems.

We show that these challenges can be overcome by incorporating personal utility and status as intrinsic rewards within the reward structure. With a reward structure combining personal utility, reputation, and status, the emergence of a common norm proceeds through the following stages:

- (1) *Personal Utility Optimization*: Agents initially adopt norms that maximize their own personal utility.
- (2) *Reputation Learning*: Although agents cannot directly observe their own reputation, they can evaluate the personal benefits gained from norms used by others. Through interactions, agents exchange information about the estimated reputations of peers, learning which norms are associated with higher reputations.
- (3) *Reputation Optimization*: Agents adopt the norms of peers with high reputations as reference points, aiming to increase their own reputation by emulating these influential norms. Agents whose norms become widely adopted are termed *influential agents*.
- (4) *Status Optimization*: Once recognized as influential, agents seek to reinforce their status by increasing community engagement and adopting norms that further solidify their recognition.
- (5) *Emergence of a Common Norm*: As members increasingly emulate influential agents, these agents gain insights into the preferences of the group, enabling them to adopt norms that maximize social welfare. These reciprocal interactions lead to the emergence of a socially optimal common norm.

In Section 3, we provide a concrete example to illustrate this process. This progression resembles how norms emerge when humans interact in a social group or community, making the process intuitive and easily understood. This connection to norm formation in social groups and communities suggests that such groups may indeed benefit from a reward structure similar to the one we propose. Moreover, it implies that norms emerging through this structure are likely to be (close to) socially optimal.

While the emergence of a common norm intuitively mirrors how humans interact in social settings, the mathematical model and analysis required to formalize and prove convergence to a socially optimal norm are more complex. This complexity stems from the multi-component reward

structure, which includes personal utility, reputation, and status. Each component incentivizes distinct behaviors, leading to a layered interdependence between agents' actions and decisions that would not arise with a single-component reward structure. Crucially, incorporating all three components is necessary because agents cannot directly measure their own reputation, requiring a combination of feedback mechanisms to drive socially beneficial outcomes.

In summary, the paper addresses a significant gap in the field of multi-agent systems: how autonomous agents can effectively coordinate and establish common norms in decentralized environments without the need for central authority or pre-programmed rules. This is particularly challenging when agents are self-interested and have only limited, local information. The gap lies in developing mechanisms that enable the emergence of socially optimal collective behaviors from individual, decentralized decision-making, which is crucial for the reliability and efficiency of complex distributed AI systems.

The most important contributions of this paper revolve around its novel three-component reward structure (personal utility, reputation, and status) and the theoretical framework built upon it. By demonstrating that this reward system, combined with a best-response dynamic, can lead to the emergence of welfare-optimal common norms, the paper provides a foundational solution to the coordination problem. Its potential impact is substantial, offering a blueprint for designing more robust, adaptive, and scalable multi-agent systems where agents can learn to cooperate and achieve collective goals autonomously, without explicit human intervention or centralized control. This could revolutionize applications ranging from autonomous vehicle fleets and robotic swarms to distributed resource management and large-scale AI networks.

More precisely, the key contributions of the paper are as follows:

- *Novel Reward Structure for Norm Emergence:* The paper introduces and formalizes a unique three-component reward structure (personal utility, reputation, and status) that is crucial for the emergence of common, socially optimal norms in decentralized multi-agent systems. It argues that the presence and interplay of all three components are essential for agents to converge on a shared norm, especially in overcoming challenges like inaccurate feedback in reputation systems.
- *Framework for Autonomous Coordination:* It provides a foundational framework for how autonomous agents, each with individual goals and limited information, can collectively arrive at socially optimal decisions. This addresses a critical, open challenge in multi-agent research by demonstrating a mechanism for establishing common knowledge and shared norms without the need for centralized control or explicit communication beyond the reward signals.
- *Theoretical Performance Guarantees:* The paper proposes to rigorously analyze the convergence of agents to a common norm using a game-theoretic framework, specifically through the concept of a Nash equilibrium. In addition, it provides learning algorithms, based on distributed stochastic gradient ascent, that enables agents to learn and optimize their local rewards. We show that the resulting dynamic leads to the emergence of a common norm that aims to maximize the overall social welfare.
- *Winner-Takes-All Effect:* The proposed reward system gives rise to a winner-takes-all effect, in which agents with slightly higher reputation or status quickly attract the attention and imitation of others. This positive feedback loop not only facilitates the emergence of a common norm—by encouraging agents to align their behavior with that of the most reputable or influential peers—but also significantly accelerates the convergence process, enabling rapid coordination in distributed systems. Importantly, under our proposed reward structure, the norm that emerges through this winner-takes-all dynamic is not arbitrary: we

show that it is (close to) socially optimal, maximizing or nearly maximizing overall social welfare. In doing so, our model provides a principled mechanism by which self-interested agents, through local interactions and social feedback, can autonomously achieve both consensus and collective efficiency without the need for centralized control.

- *Connection to Human Social Group Dynamics:* More broadly, the overall dynamic of our model — involving reputation accumulation, status-seeking behavior, the emergence of opinion leadership, and the role of social feedback mechanisms akin to gossip — strikingly mirrors patterns observed in human social groups. In real-world communities, agents are conscious of their reputation and status and adapt their behavior to enhance both. Moreover, gossip plays a central role, as individuals discuss and assess the actions and impact of others' behavior. Our model and analysis provide a formal explanation of these mechanisms: we show that a reward structure based on reputation and status, combined with gossip to evaluate the standing of others, leads to a desirable outcome — namely, the emergence of a common norm that maximizes social welfare. This connection opens the door to new interdisciplinary research at the intersection of sociology and AI, and offers a promising foundation for understanding and designing the interaction of social and artificial agent systems.

In essence, the paper offers a robust theoretical and practical framework for solving the problem of decentralized norm formation, which is paramount for the future development and deployment of complex high-stakes applications of multi-agent systems, including autonomous vehicle fleets and robotic swarms, and distributed AI networks.

In addition, the proposed progression of norm emergence, moving from individual utility to reputation and then status, mirrors the evolution of norms in human societies. This connection provides valuable insights into how complex collective behaviors can arise from simple, local interactions, potentially bridging concepts between artificial intelligence and social science.

2 PROBLEM FORMULATION

In multi-agent systems (MAS), effective coordination among agents operating within a shared environment is essential for achieving system-wide goals. Across real-world applications—from collaborative robotics and autonomous vehicle fleets to distributed sensor networks and social media platforms—agents must make decisions that balance optimizing their individual rewards with contributing to collective outcomes. A central requirement for effective coordination is for agents to establish a *norm*: a shared understanding of the environment and appropriate actions within it. This shared knowledge allows agents to synchronize their actions and work toward maximizing the overall performance of the system.

However, the challenge does not end with establishing a common norm. Once a norm is developed, agents face the additional task of selecting the shared understanding that yields the highest collective benefit. Given that agents may have heterogeneous objectives and rewards, this selection has a direct impact on both individual and system-wide outcomes. Thus, the goal is to select a norm that maximizes social welfare—defined as the total expected reward across all agents in the system.

2.1 Model Components

To formalize this problem, we introduce a model that captures the essential elements of agent coordination in a shared environment. These components reflect the dynamics of how norms emerge, how agents interact, and how they seek to optimize collective outcomes. The detailed mathematical formulation is provided in Section 2.2.

- **Common Environment:** We consider a system comprising N agents that operate within a shared environment, characterized by a finite number of states. States can represent different situations or conditions that can occur in the environment.
- **Role of Agents:** Agents can take on two distinct roles: *actor*, when performing actions directed at others, and *participant*, when observing or responding to others' actions. Different agents may allocate different amounts of time to each role.
- **Agents' Behavior as Actors:** In each state of the environment, agents decide on which action to perform. The actions they choose characterize the agent's behavior.
- **Agents' Perceived Benefit as Participants:** Agents in the participant role perceive a benefit from each observed action of another agent, which can be positive if the action is advantageous or negative if it is harmful.
- **Heterogeneous Set of Agents:** Different agents in the participant role may perceive different benefits from the same action in a given state.
- **Agents' Interactions:** Different agents in the participant role may perceive different benefits from the same action in a given state.
- **Norm:** If all agents, in their actor role, adopt the same behavior, this behavior constitutes the common norm of the group. A norm can represent not only specific actions but also rules or beliefs that agents adopt in each state.
- **Maximizing Social Welfare:** Ideally, agents identify and adopt a norm that maximizes social welfare across the entire group.

2.2 Mathematical Model

We formally represent these model components as follows.

2.2.1 Shared Environment. We consider a set of agents $\mathcal{A} = \{1, 2, \dots, N\}$, with $N \geq 2$, that operate in a shared environment characterized by a finite set of states $\mathcal{S}$. Each state $s \in \mathcal{S}$ occurs with probability $p(s)$.

In each state $s \in \mathcal{S}$, agents select an action from a finite set $\mathcal{X}_s$ of K_s possible actions. Note in each state s , that all agents choose actions from the same set of possible actions.

2.2.2 Role of Agents. Let $\mu_{a,i}$ be the rate with which agent takes on the role as an actor in the community, and $\mu_{p,i}$ the rate with which i takes on the role as a participant.

2.2.3 Agents' Behavior as Actors. The behavior of agent i is defined by a mapping $\pi_i : \mathcal{S} \rightarrow \mathcal{X}_s$, where $\pi_i(s)$ denotes the action chosen by agent i in state s . Let Π denote the set of all possible behaviors (mappings from states to actions).

2.2.4 Perceived Benefit as Participant. Agents are heterogeneous: agent $i \in \mathcal{A}$ receives a benefit $u_i(s, x)$ from action $x \in \mathcal{X}_s$ in state $s \in \mathcal{S}$. The benefit can differ across agents, states, and actions.

2.2.5 Interactions As Actors. Let $\mu_{a,i} \geq 0$ denote the rate at which agent i interacts within the group. Define the vector

$$\mu_a = (\mu_{a,1}, \dots, \mu_{a,N})$$

for all agents' rates. The probability that agent i is active within the group at a given time is modeled by a function $\theta(\mu_{a,i}) : [0, \infty) \rightarrow [0, 1]$.

2.2.6 Interactions As Participant. Let $\mu_{p,i} \geq 0$ denote the rate at which agent i interacts within the group. Define the vector

$$\mu_p = (\mu_{p,1}, \dots, \mu_{p,N})$$

for all agents' rates. The probability that agent i is active within the group at a given time is modeled by a function $\theta(\mu_{p,i}) : [0, \infty) \rightarrow [0, 1]$.

2.2.7 Norm. If all agents adopt the same behavior π , i.e.,

$$\pi_i(s) = \pi(s), \quad \forall i \in \mathcal{A}, s \in \mathcal{S},$$

then π is the (common) norm of the group.

Ideally, agents identify, and adopt, a norm π^* that maximizes social welfare given by

$$\pi^* = \arg \max_{\pi \in \Pi} \sum_{i \in \mathcal{A}} \theta(\mu_{p,i}) U_i(\pi),$$

where

$$U_i(\pi) = \sum_{s \in \mathcal{S}} p(s) u_i(s, \pi(s)).$$

2.3 Research Questions

For this setup, we study the following key questions that address both the dynamics of norm formation and its implications for system-wide outcomes:

- (1) *Norm Emergence:* How, and under what conditions, does a common norm π emerge in a fully distributed manner, based solely on local interactions between agents?
- (2) *Welfare-Optimal Norm:* Under what conditions does the emerging norm correspond to the welfare-optimal norm π^* that maximizes social welfare?
- (3) *Mechanisms and Algorithms:* Can we design distributed mechanisms or learning dynamics that reliably guide agents toward welfare-optimal norms?

2.4 Discussion

Our formulation focuses on the study of fully distributed multi-agent systems, in which agents interact locally and make decisions independently. Specifically, we are interested in settings where each agent aims solely to maximize its own reward, without centralized coordination or access to global information. The collective behavior that emerges from these local decisions determines whether the system as a whole converges to a norm.

A central question we explore is whether such decentralized dynamics can lead not only to the emergence of a common norm, but to the emergence of a norm that is *welfare-optimal* — that is, a norm that maximizes the social welfare across all agents. This mirrors dynamics seen in market economies, where individual buyers and sellers, each with their own preferences and objectives, act to maximize their own utility (buyers) or profit (sellers). Despite the absence of central planning, their interactions can produce market equilibria that are socially efficient, a phenomenon famously described as the “invisible hand” of the market.

In our setting, we ask: can we design a reward or incentive structure under which agents, by individually maximizing their own rewards in a fully distributed manner, collectively converge to a norm that maximizes social welfare? We seek conditions under which the self-interested actions of heterogeneous agents align with the collective objective of welfare maximization. Our work aims to characterize these conditions, and to develop distributed algorithms that enable such alignment without requiring centralized control.

3 REWARD STRUCTURE AND OPTIMIZATION

To create a mechanism that aligns individual agent incentives with the overall performance of the system—and leads to the emergence of a common norm that maximizes social welfare—we introduce two key components:

- (1) *Reward Structure*: We propose an incentive scheme designed to guide agents toward a welfare-optimal norm. This scheme combines three core rewards: *personal utility*, *reputation*, and *status*.

We begin by providing the intuition behind these three rewards through a concrete example, and then present their formal definitions.

3.1 Illustrative Example

While the mechanism of using three distinct core rewards may at first appear complex, its underlying intuition is simple and mirrors familiar reward systems in social groups. In the following, we highlight this connection using a simple example that not only illustrates the reward system but also the interactions and dynamics that lead to the emergence of a common norm. This example is intended to illustrate the concepts rather than serve as a practical case study.

The example illustrates:

- (1) the three types of rewards: personal utility, reputation, and status,
- (2) how agents optimize these rewards through interactions with other agents,
- (3) how reputation learning and social support play important roles in helping agents optimize their rewards,
- (4) how members continuously alternate between two roles: as *actors*, when they perform social actions directed at others, and as *participants*, when they observe or receive actions from others,
- (5) and how this reward system and these interactions lead to the emergence of a common norm.

We use this reward structure, interactions, and feedback mechanisms (reputation learning and social support) in our analysis.

Consider a gardening group in which members cultivate vegetables and flowers using diverse practices—such as planting techniques, watering schedules, and pest control methods. Members constantly shift between the roles of actors and participants: in some moments, they perform actions that others observe, and in other moments, they are participants who witness or receive the actions of others. The three types of rewards, and how agents pursue them in this group, are illustrated as follows:

- (1) *Personal Utility Optimization (Actor role)*: Alice, acting as an experimenter in the group, tries different fertilizers for her plants. Through trial and error, she discovers that a standard fertilizer works best for her. Here Alice is an actor, while the rest of the group may observe her experimentation in their role as participants.
- (2) *Reputation Learning (Participant role)*: In conversations with Bob, Alice hears about Carol, a gardener well-known for her thriving garden and eco-friendly practices. In this role, Alice is primarily a participant, receiving information and updating her perception of Carol's reputation. Carol's reputation itself reflects her past actions as an actor that were observed and valued by many participants.
- (3) *Reputation Optimization (Actor → Participant feedback loop)*: Inspired by Carol's standing, Alice seeks her advice. Carol, now acting, recommends an organic fertilizer and eco-friendly techniques. Alice adopts these practices, and as her garden flourishes, participants in the

group notice. Their recognition feeds back into Alice's reputation, showing how actions in the actor role become reputational signals to others in the participant role.

- (4) *Social Support (Participants acting back)*: As Carol's reputation grows, more members approach her for guidance. They not only observe and adopt her behavior as participants but also act in turn by praising, endorsing, and sharing her advice with others. This feedback—participants rewarding Carol—constitutes the *social support* that signals and amplifies Carol's status.
- (5) *Status Optimization (Actor reinforced by participants)*: Carol interprets the social support she receives as evidence of her elevated status. In her actor role, she responds by becoming even more attentive and generous with her expertise, which further strengthens the esteem and deference she receives from the group in their participant roles. This iterative cycle reinforces her central status.
- (6) *Emergence of a Common Norm (Mutual reinforcement of actor and participant roles)*: As more members follow Carol's techniques and provide feedback on what works best for them, her methods diffuse widely. The interplay of actor contributions and participant feedback gradually consolidates into a shared set of gardening practices, establishing a common norm that benefits the entire group.

Through these interactions, members not only maximize their individual rewards but also, in their alternating roles as actors and participants, contribute to the establishment of shared practices. The combination of visible actions and audience feedback gradually aligns behaviors, leading to the emergence of beneficial community norms.

In the following sections, we formalize this intuition and analytically demonstrate how this reward structure, combined with interactions among agents, drives the emergence of common norms that optimize social welfare.

3.2 Reward Structure

We define three core rewards that serve as an incentive mechanism for agents:

3.2.1 Personal Utility: The personal utility quantifies how beneficial a given behavior π_i is to agent i . Formally, it is defined as

$$U_i(\pi_i) = \sum_{s \in S} p(s) u_i(s, \pi_i(s)), \quad (1)$$

where $p(s)$ denotes the probability of state s , and $u_i(s, \pi_i(s))$ is the utility agent i derives from action $\pi_i(s)$ in state s . When maximizing its personal utility, agent i selects the behavior that is most beneficial to itself.

3.2.2 Reputation: The reputation of an agent characterizes how beneficial the behavior of agent i is to other agents. Formally, it is defined as

$$R_i(\pi_i) = \sum_{j \in C \setminus \{i\}} \theta(\mu_{a,i}) \theta(\mu_{p,j}) U_j(\pi_i). \quad (2)$$

The terms $\theta(\mu_{a,i})$ and $\theta(\mu_{p,j})$ weight the reputational effect of agent i 's behavior by (i) how often i 's actions are visible as an actor, and (ii) how much attention and evaluative weight participant j assigns to others' actions. Thus, reputation depends not only on the objective benefits conferred but also on the visibility of actors and the attentiveness of audiences.

The perceived reward (value) from this reputation is given by

$$\gamma R_i(\pi_i),$$

where $\gamma > 0$, and $U_j(\pi_i)$ is the benefit that agent i 's behavior provides to agent j . The constant γ represents the weight that agent i places on adopting behaviors that benefit others and thereby increase its reputation.

3.2.3 Status and Social Support: The status of an agent k is characterized by how many agents follow k and adopt its behavior, as well as by the benefit these agents derive from doing so. Formally, it is defined as

$$S_k(\pi_k) = \sum_{i \in \mathbb{F}_k} \theta(\mu_{p,i}) R_i(\pi_k). \quad (3)$$

where $R_i(\pi_k)$ is the reputation that agent i obtains by adopting agent k 's behavior.

We interpret

$$\theta(\mu_{p,i}) R_i(\pi_k)$$

as the social support of agent i to agent k . Here, "social support" refers to the symbolic endorsement or recognition an agent gains when others adopt their behavior. Each follower $i \in \mathbb{F}_k$ provides social support to agent k proportional to the reputation $R_i(\pi_k)$ that they receive from adopting the behavior of agent k .

Agents express or communicate their social support through their adoption of agent k 's behavior, which serves as a public signal of approval and endorsement within the community. In order to adopt this behavior, followers typically interact with agent k to learn and observe it, during which they provide positive feedback proportional to the reputation they gain from agent k 's behavior. This positive feedback can take various forms, including verbal acknowledgment and other supportive actions. The status thus represents the total reward that agent k perceives from the social support of its followers.

The perceived reward from status is given by

$$\kappa S_k(\pi_k),$$

where $\kappa > 0$ is the status parameter. The constant κ represents the weight that agents place on having their behavior adopted by others.

3.3 Role Optimization

In the following, we assume that agents focus on maximizing the reward component (personal utility, reputation, or status) that provides them with the highest payoff at any given time. This leads to the following dynamic interaction: when the reputation of an agent k increases, it attracts more followers who adopt its behavior, thereby boosting both its status and further reinforcing its reputation. This creates a positive feedback loop, akin to a winner-takes-all effect, where an agent with even a slightly higher reputation than others may ultimately attract all other agents as followers, achieving a reputation and status far greater than any other agent. Our analysis formalizes this intuition.

3.4 Reward Optimization

3.4.1 Optimizing Personal Utility. To maximize its personal utility, agent i selects a behavior π_i such that

$$\pi_i = \arg \max_{\pi \in \Pi} \sum_{s \in S} p(s) u_i(s, \pi(s)).$$

3.4.2 Reputation Learning and Optimization. The reputation of an agent characterizes how beneficial the behavior of agent i is to other agents. Formally, it is defined as

$$R_i(\pi_i) = \sum_{j \in \mathbb{C} \setminus \{i\}} \theta(\mu_{a,i}) \theta(\mu_{p,j}) U_j(\pi_i). \quad (4)$$

Note that agent i must interact with others and gather information to identify which agent k has the highest reputation in the group. We refer to this process as *reputation learning*.

To maximize its reputation, agent i identifies the agent k^* with the highest observed reputation:

$$k^* = \arg \max_{k \in \mathcal{A}} R_k(\pi_k),$$

and adopts the behavior of agent k^* , i.e.,

$$\pi_i = \pi_{k^*}.$$

3.4.3 Status Optimization. To maximize its status, agent k selects a behavior π_k such that

$$\pi_k = \arg \max_{\pi \in \Pi} \sum_{i \in \mathbb{F}_k} \theta(\mu_{p,i}) R_i(\pi_k)$$

Note that this is equivalent to choose a behavior π_k that maximizes agent's k reputation, and

$$\pi_k = \arg \max_{\pi \in \Pi} R_i(\pi).$$

3.5 Optimizing Interaction Rates as Actor

Agent i optimizes its interaction rate as an actor as follows.

Let π_i be the current behavior of agent i , and let

$$R_{\max} = \max_{i \in \mathcal{A}} R_i(\pi_i)$$

be the highest reputation over all agents. Furthermore let H_i be agent i 's maximum reward over the different roles given by

$$H_i = \max\{U_i(\pi_i), \gamma R_i(\pi_i), \kappa S_i(\pi_i)\}.$$

Finally, let u_0 be the reward an agent obtains from interactions outside the group.

Given a total interaction rate budget $M > 0$, agent i chooses its interaction rate μ_i , and its interaction rate λ_i outside the group, using the following optimization problem,

$$\begin{aligned} \max_{\lambda_i, \mu_{a,i} \geq 0} \quad & \theta(\lambda_{a,i}) u_0 + \theta(\mu_{a,\min} + \mu_{a,i}) H_i \\ \text{s.t.} \quad & \lambda_{a,i} + \mu_{a,i} \leq M. \end{aligned} \tag{5}$$

3.6 Discussion

In the formulation above, we assume that agents observe the benefits of others' actions accurately and update their evaluations without error. This assumption ensures that the definitions of reputation and status are consistent with realized payoffs and allows us to conduct a clean game-theoretic analysis of equilibrium behavior. Introducing imperfect observation at this stage would complicate the analysis by conflating incentive effects with noise in information processing.

Later, when we consider learning dynamics, we relax this assumption and allow agents to form reputational assessments based on noisy or incomplete information. In that setting, agents must update beliefs under uncertainty, and reputational signals may diverge from actual contributions. This extension provides a more realistic model of distributed systems where perception and evaluation are inherently imperfect.

To simplify the analysis and notation, we assume that agents have fixed interaction rates as participants. These interaction rates could also be treated as decision variables, similar to the optimization of interaction rates by an actor. Doing so does not change the fundamental results

or insights presented in this paper, but it would require a more general definition of the welfare-optimal norm and would complicate both the Nash equilibrium analysis in Section ?? and the analysis of the learning algorithms in Sections ??–??.

4 ANALYTICAL FRAMEWORK

Under the proposed reward structure, agents choose to optimize the reward component—personal utility, reputation, or status—that provides the highest payoff. In this section, we provide the framework that we use for the analysis. Specifically, we define the key concepts, notations, and roles that describe agent decisions and interactions. We then outline two complementary analytical approaches: a game-theoretic analysis assuming perfect information, and a learning-based analysis for distributed, information-limited environments. Finally, we specify the fundamental assumptions on which our analysis relies. Together, these elements allow us to characterize when and under which conditions common norms emerge under our proposed incentive structure—and whether such norms align with social welfare.

4.1 Assumptions

For our analysis—both the Nash equilibrium analysis and the analysis of the learning algorithms—we make the following three assumptions.

Our first assumption requires that the personal utilities $U_i(\pi)$ of all agents $i \in \mathcal{A}$ are bounded.

ASSUMPTION 1. *There exists a bound B_U such that*

$$|U_i(\pi)| \leq B_U, \quad i \in \mathcal{A}, \pi \in \Pi.$$

Assuming that the personal utilities of all agents are bounded by a constant B is a standard and mild assumption commonly adopted in game-theoretic and learning analyses. Bounded utilities ensure that payoffs remain finite, which is essential for guaranteeing the mathematical well-posedness of the model and for applying convergence results in learning algorithms. Moreover, this assumption aligns well with practical scenarios where utilities represent measurable quantities that naturally have upper and lower limits. While it may exclude some theoretical models with unbounded payoffs, it does not significantly restrict the applicability of our results in the context of norm emergence and strategic interactions among agents.

The next assumption concerns a given group $\mathcal{A}$ of agents. We assume that for at least one agent, the behavior π_i^{PU} that maximizes personal utility also leads to a positive reputation.

ASSUMPTION 2. *There exists an agent $i \in \mathcal{A}$ such that*

$$\sum_{j \in C \setminus \{i\}} U_j(\pi_i^{PU}) > 0.$$

Assumption 2 is mild and reasonable, as it simply ensures that individual incentives and social evaluations are not entirely misaligned for all agents. In most social or multi-agent systems, it is natural to expect that some behaviors that maximize personal utility (e.g., cooperation or fairness) also generate positive social feedback. This assumption rules out pathological cases where every personally optimal action is purely antisocial. However, it may fail in adversarial environments where personal gain comes only at others' expense, or in highly fragmented groups where conflicting standards prevent any action from receiving positive reputation. In such settings, norm emergence driven by reputation would be inhibited.

4.2 Approach

Depending on which reward component an agent optimizes (that is, which set $\mathbb{P}(\Phi)$, $\mathbb{R}(\Phi)$, or $\mathbb{S}(\Phi)$ they belong to), the agent plays a different role in the group. As part of the analysis, we characterize the dynamics of how agents switch roles, the resulting decisions of each agent, and whether this process leads to the emergence of a common norm—and ideally to the emergence of a common norm that maximizes social welfare. To do this, we proceed as follows.

We first analyze this process using a game-theoretic framework, where we assume that each agent has perfect knowledge of the current decisions of all other agents, and uses this perfect information to decide which reward component to optimize. For this setup, we characterize when there exists a Nash equilibrium that leads to the emergence of a common norm, and when there exists a Nash equilibrium that leads to the emergence of a common norm that maximizes social welfare. This Nash equilibrium analysis provides fundamental insights into whether and under which conditions a common norm emerges under our proposed reward structure.

The assumption that agents have perfect information, which we use for the Nash equilibrium analysis, is unrealistic for practical settings. To address this, we next consider the case where agents do not have a priori perfect information, but can acquire information through interaction with other agents. This leads to learning algorithms in which agents iteratively learn (estimate) the payoffs under the different reward components and optimize the component that currently yields the highest payoff. We again characterize for these learning algorithms whether and under which conditions a common norm emerges under our proposed reward structure. This setup yields realistic algorithms suitable for practical situations where agents must make decisions and learn in a fully distributed environment.

5 NASH EQUILIBRIA, OPINION LEADERS, AND WELFARE-OPTIMAL NORMS

In this section we carry out the Nash equilibrium analysis where we study whether agents pursuing individual goals within the proposed reward structure eventually adopt a common norm π^* that maximizes social welfare. For this analysis we assume that agents have *perfect information*: each agent knows the role, rate allocation, and behavior of every other agent in the system. We then define a Nash equilibrium as follows:

5.1 Notation and Definitions

For this, we use the following notation and definitions.

5.1.1 Group Configuration: Agent Decisions and Followers. Each agent $i \in \mathcal{A}$ must decide on its behavior π_i , its interaction rate μ_i within the group, and its interaction rate λ_i outside the group. Let

$$\Phi_i = (\pi_i, \mu_{a,i}, \lambda_{a,i}, \mu_{p,i}, \lambda_{p,i})$$

denote the decision vector of agent i , and let

$$\Phi = (\Phi_1, \dots, \Phi_N)$$

denote the decision vector across all agents $i \in \mathcal{A}$.

In addition, let

$$\mathbb{F} = (\mathbb{F}_1, \dots, \mathbb{F}_N)$$

be the vector indicating the set of followers of each agent.

Finally, let

$$\Omega = (\Phi, \mathbb{F})$$

denote the configuration of the group, including the decisions and the follower sets of all agents.

5.1.2 Behavior that Maximizes Personal Utility. For each agent $i \in \mathcal{A}$, let π_i^{PU} be the behavior π_i^{PU} that maximizes its personal utility $U_i(\pi_i)$, given by

$$\pi_i^{PU} = \arg \max_{\pi_i \in \Pi} \sum_{s \in S} p(s) u_i(s, \pi(s)).$$

Note that π_i^{PU} does not depend on the decisions of other agents, but only on the preferences of agent i .

5.1.3 Agents with the Highest Reputation. Furthermore, given a the decision vector $\Phi = (\Phi_1, \dots, \Phi_N)$, let $\mathcal{A}^*(\Phi)$ be the set of agents with the highest observed reputation, i.e.,

$$k^* = \arg \max_{i \in \mathcal{A}} R_i(\pi_i), \quad k^* \in \mathcal{A}^*(\Phi).$$

5.2 Agents' Role

Using these definitions, agents choose which reward component to optimize as follows for a given configuration $\Omega = (\Phi, \mathbb{F})$.

5.2.1 Agents that Optimize Reputation. All agents $i \notin \mathcal{A}^*(\Phi)$ decide whether to optimize their personal utility or their reputation. They will choose to maximize their reputation by adopting the behavior of an agent $k^* \in \mathcal{A}^*(\Phi)$ if

$$R_{k^*}(\pi_{k^*}) > B_R$$

and

$$\gamma R_i(\pi_{k^*}) > U_i(\pi_i^{PU}),$$

where

$$U_i(\pi_i^{PU}) = \sum_{s \in \mathcal{S}} p(s) u_i(s, \pi_i^{PU}(s)).$$

Let $\mathbb{R}(\Omega)$ be the set of all agents $i \notin \mathcal{A}^*(\Phi)$ for which this condition holds.

To optimize its reputation, an agent $i \in \mathbb{R}(\Omega)$ follows an agent $k^* \in \mathcal{A}^*(\Phi)$ and adopts its behavior. If $\mathcal{A}^*(\Phi)$ contains more than one agent, agent i selects an agent k^* uniformly at random from $\mathcal{A}^*(\Phi)$.

5.2.2 Agents that Optimize Status. All agents $k^* \in \mathcal{A}^*(\Phi)$ decide whether to optimize their personal utility or their status. They will choose to maximize their status if their observed status exceeds the payoff from personal utility maximization, i.e.,

$$\kappa S_{k^*}(\pi_{k^*} | \mathbb{F}_{k^*}) > U_{k^*}(\pi_{k^*}^{PU}).$$

Let $\mathbb{S}(\Omega)$ be the set of all agents $k^* \in \mathcal{A}^*(\Phi)$ for which this condition holds.

5.2.3 Agents that Optimize Personal Utility. All agents $i \in \mathcal{A}$ that do not optimize reputation (i.e., $i \notin \mathbb{R}(\Omega)$) and do not optimize status (i.e., $i \notin \mathbb{S}(\Omega)$) optimize their personal utility. Let $\mathbb{P}(\Omega)$ be the corresponding set of agents.

5.3 Nash Equilibrium

DEFINITION 5.1. A configuration $\Omega^* = (\Phi^*, \mathbb{F}^*)$ with

$$\Phi^* = (\Phi_1^*, \dots, \Phi_N^*) \quad \text{and} \quad \Phi_i^* = (\pi_i^*, \mu_i^*, \lambda_i^*),$$

and

$$\mathbb{F}^* = (\mathbb{F}_1^*, \dots, \mathbb{F}_N^*),$$

is a Nash equilibrium if the following holds:

(1) For all agents $i \in \mathbb{P}(\Omega^*)$ we have

$$\pi_i^* = \pi^{PU} i$$

and

$$(\mu_i, \lambda_i) = \arg \max_{\lambda, \mu} [\theta(\lambda) u_0 + \theta(\mu) \pi^{PU} i].$$

(2) For all agents $i \in \mathbb{R}(\Omega^*)$, there exists $k^* \in \mathcal{A}^*(\Phi^*)$ such that

$$i \in \mathbb{F}_{k^*}^* \quad \text{and} \quad \pi_i^* = \pi_{k^*}^*,$$

and

$$(\lambda_i^*, \mu_i^*) = \arg \max_{\lambda, \mu} [\theta(\lambda) u_0 + \theta(\mu) \hat{J}_{k^*}(\pi_{k^*}^*, \mu_{k^*}^*)].$$

(3) For all agents $i \in \mathbb{S}(\Omega^*)$ we have

$$\Phi_i^* = \arg \max_{\Phi} [\theta(\lambda) u_0 + \theta(\mu) \hat{J}_i(\pi, \mu | \mathbb{F}_i^*)].$$

By definition, under a Nash equilibrium Ω^* , no agent has an incentive to unilaterally change its decision—that is, its behavior, rate allocation or role. Consequently, a stable configuration is reached.

In the remainder of this section, we study the conditions under which a Nash equilibrium Ω^* with a common norm

$$\pi_i^* = \pi^*, \quad \forall i \in \mathcal{A},$$

emerges, and whether this common norm maximizes social welfare.

5.4 Sufficient Condition for The Emergence of a Common Norm π

We first provide a sufficient condition that guarantees the existence of a Nash equilibrium with a common norm. We have the following result.

LEMMA 1. *Given a Nash equilibrium Φ^* , if there exists an agent $k^* \in \mathcal{A}$ who is followed by all other agents, i.e.,*

$$\mathbb{F}_{k^*}^* = \mathcal{A} \setminus \{k^*\},$$

then $\pi_{k^}^*$ is adopted by all agents in the group and becomes the common norm. That is,*

$$\pi_i^* = \pi_{k^*}^*, \quad \forall i \in \mathcal{A}.$$

The result of Lemma 1 follows directly from the fact that, by definition, all agents $i \in \mathbb{F}_{k^*}^*$ adopt the behavior of agent k^* . We omit the detailed derivation for brevity.

We introduce the following convention. For a Nash equilibrium Ω^* satisfying the condition of Lemma 1, we refer to the corresponding agent k^* as an *opinion leader* (or *influencer*), reflecting their role as a central figure whose behavior determines the common norm of the group. Furthermore, we refer to a Nash equilibrium Ω^* as described in Lemma 1 as a Nash equilibrium with opinion leader k^* .

For a Nash equilibrium with opinion leader k^* , we introduce a slightly weaker notion of a Nash equilibrium with a common norm that maximizes social welfare.

DEFINITION 5.2. *Let Ω^* be a Nash equilibrium with opinion leader k^* and common norm $\pi^* = \pi_{k^*}^*$. We say that the common norm π^* maximizes social welfare if*

$$\pi^* = \arg \max_{\pi \in \Pi} \sum_{i \in \mathcal{A} \setminus \{k^*\}} \theta(\mu_{p,i}) U_i(\pi).$$

This definition states that we consider a Nash equilibrium with opinion leader k^* to have a common norm that maximizes social welfare if that norm maximizes the total utility of the followers of the opinion leader.

5.5 Existence of a Nash Equilibrium with Common Norm π^*

We then obtain the following result on the existence of a Nash equilibrium with a common norm π^* .

PROPOSITION 1. *Let Assumption 2 hold. Given N , M , and $\kappa = 0$, there exists a $\gamma_0 > 0$ such that for all $\gamma > \gamma_0$, a Nash equilibrium with an opinion leader k^* and a common norm π^* exists.*

We provide a proof of Proposition 1 in Appendix ???. The key idea behind the proof is that a sufficiently large γ amplifies the reward for reputation to the point that agents achieve a higher overall reward by maximizing their reputation rather than their personal utility. As a result, all agents become followers and adopt the norm of the agent k^* with the highest reputation, who thereby becomes the opinion leader.

Having established conditions for the existence of a Nash equilibrium with a common norm, we next provide conditions for the existence of a Nash equilibrium with a common norm π^* that maximizes social welfare.

PROPOSITION 2. *Let Assumption 2 hold. Given N and M , there exist $\gamma_0 > 0$ and $\kappa_0 > 0$ such that for all $\gamma > \gamma_0$ and $\kappa > \kappa_0$, there exists a Nash equilibrium with an opinion leader k^* and a common norm π^* that maximizes social welfare as given in Definition 5.2.*

We provide a proof of Proposition 2 in Appendix ??.

Proposition 2 extends Proposition 1 to the case of a welfare-optimal norm, introducing an additional condition on the parameter κ . The key idea behind the proof is that if the incentive parameter κ for optimizing status is sufficiently large, agent k^* will choose to optimize its status, as this provides a higher reward than maximizing personal utility. Furthermore, if γ is sufficiently large, all other agents will prefer to maximize their reputation by following agent k^* rather than maximizing their own personal utility. As a result, there exists a Nash equilibrium with a single opinion leader k^* as described in Lemma 1, and this opinion leader optimizes its status, leading to a common norm that maximizes social welfare.

5.6 Convergence to a Nash Equilibrium with the Optimal Common Norm π^*

The results we have obtained provide conditions under which a Nash equilibrium with a common norm that maximizes social welfare exists. But how likely is it that this norm will actually emerge as agents dynamically adapt their behavior, role, and rate allocation based on the actions of other agents? Here, we analyze this question in the context of the *best-response dynamic*.

The best-response dynamic consists of collective strategies that are improved through iterative steps, where each agent updates its strategy based on the strategies of the other agents in the previous iteration. Essentially, the best-response dynamic provides a natural mechanism by which agents choose strategies that maximize their individual benefit at each iteration.

Formally, the best-response dynamic generates a sequence of decision vectors,

$$\Omega_1, \Omega_2, \Omega_3, \dots$$

where, given an initial decision vector Ω_1 , each subsequent configuration Ω_n for $n \geq 2$ is obtained as follows: each agent $i \in C$ updates its decision by maximizing its reward (as outlined in Sections 3.4.1–5.2), using the configuration Ω_{n-1} from the previous iteration as input.

We make the following assumption on the initial configuration Ω_1 for our analysis.

ASSUMPTION 3. *For the initial configuration $\Omega_1 = (\Phi_1, \mathbb{F}_1)$ with*

$$\Phi_1 = (\Phi_{1,1}, \dots, \Phi_{1,N}) \quad \text{and} \quad \Phi_{1,i} = (\pi_{1,i}, \mu_{1,i}, \lambda_{1,i}),$$

and

$$\mathbb{F}_1 = (\mathbb{F}_{1,1}, \dots, \mathbb{F}_{1,N}),$$

we have that

$$\pi_{1,i} = \pi_i^{PU}$$

and

$$(\mu_{1,i}^*, \lambda_{1,i}^*) = \arg \max_{\lambda, \mu} [\theta(\lambda) u_0 + \theta(\mu) \pi^{PU} i].$$

Furthermore, we have that

$$\mathbb{F}_{1,i} = \emptyset.$$

The assumption on the initial configuration states that, at the outset, all agents optimize their personal utility. This is both a mild and practical assumption. In the early stages of interaction, when agents have no information about the behavior, preferences, or strategies of others, it is reasonable for them to act in a way that maximizes their own utility. Such behavior reflects the natural starting point for decision-making in the absence of social knowledge or coordination, and provides a neutral baseline from which patterns of interaction, learning, and adaptation can emerge over time. Algorithm 1 describes the resulting best-response dynamic.

ALGORITHM 1: Best-Response Dynamic**Data:** N, M, γ, δ , and $u_i(s, x)$ for all states $s \in \mathcal{S}$ and actions $x \in \mathcal{X}_s$ in state s **Result:** A sequence of allocations $\Phi_1, \Phi_2, \dots$, which converge to a Nash equilibrium Φ^* with the optimal common norm π^* $n = 1$;**foreach** $i \in C$ **do** $(\pi_i, \mu_i, i) \leftarrow$ optimal strategy given agent i is self-interested ; $\Phi_{1,i} \leftarrow (\pi_i, \mu_i, i)$;**end** $\Phi_1 \leftarrow (\Phi_{1,i})_{i \in C}$;**while** *True* **do** $n = n + 1$; **foreach** $i \in C$ **do** // Computations based on the allocations (π_j, μ_j, h_j) for all agents $j \neq i$ $(U, R, S) \leftarrow$ optimal reward based on each role agent i can be; $\text{role}i' \leftarrow \arg \max(U, R, S)$; $(\pi'_i, \mu'_i, \text{role}i') \leftarrow$ optimal strategy for agent i given it has the role $\text{role}i'$; $\Phi_{n,i} \leftarrow (\pi'_i, \mu'_i, \text{role}i')$; **end** $\Phi_n \leftarrow (\Phi_{n,i})_{i \in C}$; $(\pi_i, \mu_i, \text{role}i) \leftarrow (\pi'_i, \mu'_i, \text{role}i')$;**end**

We say that the best-response dynamic converges to the configuration Ω^* , if we have that

$$\lim_{n \rightarrow \infty} \Omega_n = \Omega^*.$$

We then have the following result.

PROPOSITION 3. *Let Assumption 2 and Assumption 3 hold. Given N and M , there exist $\gamma_0 > 0$ and $\kappa_0 > 0$ such that for all $\gamma > \gamma_0$ and $\kappa > \kappa_0$, the best-response dynamic converges to a Nash equilibrium with an opinion leader k^* and a common norm π^* that maximizes social welfare as given by Definition 5.2.*

We provide a proof of Proposition 3 in Appendix ??.

Proposition 3 states that for sufficiently large γ and κ , the best-response dynamic converges to a Nash equilibrium with a common norm. While the conditions for convergence to a Nash equilibrium with an optimal common norm are the same as the conditions for existence provided in Proposition 2, the convergence may require a higher thresholds γ_0 and κ_0 for γ and κ .

5.7 Discussion

The Nash equilibrium analysis offers novel insights to the literature on norm emergence and welfare-optimal consensus in multi-agent systems. While prior work on social learning, network games, and evolutionary models of norms (e.g., [?], [?], [?]) has examined conditions for consensus or convergence to stable behaviors, these models typically do not provide explicit conditions under which the emergent norms are welfare-optimal. In contrast, we derive precise Nash equilibrium conditions linking group size, social feedback strength, and incentive parameters (reputation and status) to the emergence of shared norms that maximize collective welfare. Our procedural model uniquely integrates dynamic social feedback processes (such as imitation and evaluation) with

formal equilibrium analysis, bridging a gap between evolutionary and game-theoretic approaches. Moreover, by identifying the star network structure as the ideal configuration for promoting welfare-optimal norm convergence, our results formalize and extend longstanding intuitions from social theory (e.g., on charismatic authority and focal agents) in a rigorous equilibrium framework. To our knowledge, no prior work combines these elements to provide explicit welfare conditions for norm emergence in this way. Concretely, our main contributions are as follows:

- (1) **Existence of common-norm equilibria under broad conditions:** We rigorously characterize conditions (e.g., sufficiently large populations, adequate budget rates, and social preference parameters) under which Nash equilibria with common norms exist. We show that these conditions are not restrictive, highlighting that norm convergence is not a fragile or rare phenomenon.
- (2) **Emergence of welfare-optimal norms:** Our analysis shows that, under the identified conditions, not only do agents converge to a shared norm, but the norm maximizes social welfare. This shows that individual strategic behavior can yield socially desirable outcomes, addressing a fundamental tension in game-theoretic models of norm formation.
- (3) **Critical role of social preferences and cooperation:** The parameters κ and γ , capturing the value agents place on information sharing and on norms that benefit others, are shown to be decisive for achieving both norm convergence and welfare maximization. This highlights the importance of fostering cooperative incentives in the design of multi-agent systems.
- (4) **Role of group size N :** The analysis provides the interesting and somewhat surprising result that the larger the group size, the easier it becomes for a common norm—and one that maximizes social welfare—to emerge.
- (5) **Best-response dynamics and decentralized convergence:** Our results show that a simple decentralized procedure—the best-response dynamic—leads to convergence to Nash equilibria with welfare-optimal norms. This provides a constructive and feasible pathway for norm emergence in realistic systems where agents iteratively adjust their strategies based on local observations.

How useful are these insights? Do they allow us to make progress on the fundamental question: *How can decentralized societies or systems achieve stable, welfare-maximizing norms without external enforcement or central coordination?* Taken together, the findings indeed offer a novel and significant contribution to this question. They show that endogenous dynamics—driven solely by local incentives, imitation, and reputational feedback—can produce both equilibrium stability and social optimality. Local strategic behavior, far from leading to disorder or inefficiency, can generate shared norms that align individual incentives with collective welfare. This advances our understanding of how normative behavior can arise organically in decentralized systems. It demonstrates that robust and socially desirable outcomes can emerge naturally from the interplay of individual decision-making and social feedback, without reliance on exogenous authorities, designed rules, or explicit coordination mechanisms. These insights open promising avenues for designing multi-agent systems, distributed algorithms, and institutional frameworks that harness endogenous dynamics to achieve desirable outcomes in complex, real-world environments.

In the following section, we build on these results to derive a fully distributed learning algorithm that does not require the assumption of perfect information made in our Nash equilibrium analysis.

6 LEARNING ALGORITHMS FOR PERSONAL UTILITY, REPUTATION, AND STATUS

In the previous section, we analyzed how agents can achieve socially beneficial coordination by optimizing one of three reward types: personal utility, reputation, or status. Each of these rewards corresponds to a distinct role that agents can take within the system—self-interested individual, follower, or opinion leader.

In this section, we study how agents can learn to optimize each reward type in a decentralized setting, based on limited local information and interactions with other agents. We focus separately on the learning dynamics for each role: first, how agents can learn to maximize their own personal utility; second, how they can learn to identify and imitate peers with high reputation; and third, how influential agents can learn to optimize their status by responding to social feedback.

Importantly, in this section, we assume that an agent's role is fixed—that is, each agent is committed to optimizing one of the three reward types. In the next section, we extend this analysis to how agents decide which reward to optimize.

6.1 Definitions and Notation

In this subsection we introduce additional definitions and notation that we define the learning algorithms that we consider.

6.1.1 Behavior Representation. Each agent i maintains a parameter vector $w_i \in \mathbb{R}^d$, which defines a *probabilistic behavior* $\pi(w_i; s, x)$, representing the probability of choosing action $x \in X_s$ in state s . These probabilities are computed using a softmax over a scoring function $\phi(w_i; s, x)$:

$$\pi(w_i; s, x) = \frac{\exp(\phi(w_i; s, x))}{\sum_{x' \in X_s} \exp(\phi(w_i; s, x'))}. \quad (6)$$

The function ϕ could be linear or nonlinear; we remain agnostic to its specific form in this analysis. We do however assume that the behavior representation $\phi(w; s, x)$ has the following property.

ASSUMPTION 4. *The behavior representation $\phi(w; s, x)$ is differentiable in the weights w , and its derivative $\nabla_w \phi(w; s, x)$ is Lipschitz continuous and bounded.*

6.1.2 Time-dependent Variables. In the following we consider online algorithms, where agents at each time step iteratively update the weights w_i of their policy representation, as well as their interaction rate allocations, and their roles. For this, we use the following notation.

Let $s(t)$ be the state at time t , let $w_i(t)$ the weights of agent i at time t , and let $\mu_i(t)$ and $\mu_i(t)$ be the interactions rates at time t .

Furthermore, let $\mathbb{P}(t)$, $\mathbb{R}(t)$, and $\mathbb{S}(t)$, indicate which agents optimize their personal utility, reputation, and status, respectively, at time t .

Also, let $w_i^{pu}(t)$, $w_i^r(t)$, and $w_i^s(t)$, be the weights that agent i for its behavior in the different roles. Agent i initializes these weights at time $t = 0$, and then uses the learning algorithms that we introduce in this section to update its weights for $t \geq 1$.

Finally, let $\mu_{a,i}(t)$ and $\mu_{p,i}$, be the interaction rates of agent i within the community as an actor (i.e. choosing and taking an action) and as a participant (i.e. observing actions of other agents), as function of time t . Here, we assume that the interaction rates of agents as a participant are fixed (do not change over time) and given by $\mu_{p,i}$, $i \in \mathcal{A}$, but agents update their interaction rates as an actor $\mu_{a,i}(t)$ over time as we describe below.

Using these rates, we define the probability that an agent is active at time t as an actor, or as a participant, using the function θ , we for concreteness we assume that this function is given as

follows:

$$\theta(\mu) = 1 - \exp -\mu, \quad \mu \geq 0.$$

In this section, we focus on how agents update their weights $w_i(t)$ and their interaction rates as an actor $\mu_{a,i}(t)$, in the next section with focus on how agents update their roles and the resulting dynamic of the sets $\mathbb{P}(t)$, $\mathbb{R}(t)$, and $\mathbb{S}(t)$.

Importantly, agent i updates its weight $w_i(t)$ only if agent i is active in the group at time t .

6.2 Set of Active Agents $\mathcal{A}_a(t)$ and $\mathcal{A}_p(t)$

For each time step $t \geq 1$, we define the two sets $\mathcal{A}_a(t)$ and $\mathcal{A}_p(t)$.

$\mathcal{A}_a(t)$ is the set of agents that are active as an actor at time t . Each agent i is included into this set with probability $\theta(\mu_{a,i}(t))$ independently from all other agents and choices, where

$$\theta(\mu) = 1 - \exp -\mu, \quad \mu \geq 0.$$

$\mathcal{A}_p(t)$ is the set of agents that are active as participants at time t . Each agent i is included into this set with probability $\theta(\mu_{p,i})$ independently from all other agents and choices, where

$$\theta(\mu) = 1 - \exp -\mu, \quad \mu \geq 0.$$

Recall that the the rates $\mu_{p,i}$ are fixed for all agents, and do not change over time.

6.3 Optimizing Personal Utility

We begin by defining how an agent $i \in \mathcal{A}_a(t)$ that is active in the group at time t updates its weights $w_i^{pu}(t)$ to maximize its personal utility.

6.3.1 Objective Function. In this case, the goal of agent i is to learn parameters w_i^{pu} that maximize its expected utility under the policy $\pi(w_i^{pu}; \cdot, \cdot)$:

$$J^{pu}(w_i) = \sum_{s \in \mathcal{S}} p(s) \sum_{x \in X_s} \pi(w_i; s, x) u_i(s, x), \quad (7)$$

where $u_i(s, x)$ is the utility received by agent i for taking action x in state s .

6.3.2 Gradient Estimation and Update. To optimize $J(w_i)$, the agent i is using a standard stochastic gradient method. At time t , the agent i first choose an action $x_i(t)$ where the action $x_i(t)$ randomly according to the random policy $\pi(w_i(t); s(t), x)$ for the current weight vector $w_i(t)$. Next, agent i updates its weight vector by setting

$$w_i^{pu}(t+1) = w_i^{pu}(t) + \alpha(t) \cdot u_i(s(t), x_i(t)) \cdot \nabla_w \log \pi(w_i^{pu}(t); s(t), x_i(t)), \quad (8)$$

where $\alpha(t)$ is a stepsize parameter, and $u_i(s(t), x_i(t))$ is the personal utility of agent i in state $s(t)$ and action $x_i(t)$.

This weight update increases the likelihood of actions that yield higher utility and reduces the likelihood of less effective actions.

6.4 Reputation Learning and Optimization

We now describe how each agent $i \in C$ learns to identify the agent $k^* \in C \setminus \{i\}$ with the highest observed reputation as defined in Eq. (??), and use the behavior of agent k^* to optimize its own reputation. Designing an algorithm that is both scalable and efficient in helping agents to identify an agent with the highest observed reputation is a key challenge of the overall learning algorithm, as we explain below.

To identify an agent $k^* \in C \setminus \{i\}$ with the highest observed reputation, agents must (locally) interact to exchange their estimates of the perceived benefits of others' behavior. However, this

local exchange of reputation estimates poses a scalability challenge. In large groups, agents may need to communicate a substantial number of estimates, which can result in prohibitive overhead in systems with limited communication capacity. To address this, we propose a novel algorithm in which, at each time step, agents only exchange their estimate for the agent they currently believe has the highest observed reputation. We formally show that this communication-efficient algorithm still allows agents to correctly identify the highest-reputation agent with probability 1.

More precisely, we use an approach with the following two key components. First, each agent keeps locally an estimate of the personal benefit they receive from the behavior of other agents. Second, agents interact with each other and exchange estimates of their personal benefit in order to obtain an estimate of the benefit of an agent's behavior to the overall group. This estimate of the the benefit of an agent's behavior to the overall group is equivalent to the observed reputation of an agent.

6.4.1 Local Information. To define the algorithm, we first define the information that each agent $i \in C$ keeps locally.

Recall that the reputation of an agent $k \in C$ is defined as:

$$R_k(\pi_k, \mu_k) = \sum_{i \in C \setminus \{k\}} \theta(\mu_k) U_i(\pi_k),$$

where π_k is the norm used by agent k , $U_i(\pi_k)$ is the utility agent i derives from it, and $\theta(\mu_k)$ is the probability that agent k is active in group at time t .

Each agent $i \in C$ maintains the following time-dependent local estimates at each time step $t \in \mathbb{N}$:

- (1) *Personal Benefit Estimates* $v_i(k, t)$: an estimate of the benefit $\theta(\mu_k(t)) U_i(\pi_k(t))$ that agent $k \neq i$'s norm provides to agent i .
- (2) *Observed Reputation Estimates* $s_i(k, t)$: an estimate of the reputation $R_k(\pi_k(t), \mu_k(t))$ of agent $k \neq i$.
- (3) *Estimate of Agent with the Highest Observed Reputation*: Each agent i keeps a estimate $L_i(t)$ of which agent they believe is the agent with the highest observed reputation.

We now define how each of these local estimates are initialized and updated over time.

6.4.2 Learning Personal Benefit. Each agent i initializes its estimate of the personal benefit of other agents' behavior by setting

$$v_i(k, 0) = 0 \quad k \in C \setminus \{i\}.$$

Each agent i updates its personal benefit estimates for all other agents that $k \in \mathcal{A}_a(t)$ that are active in the group at time t by setting

$$v_i(k, t+1) = [v_i(k, t) + \eta^v(t) [u_i(s(t), x_k(t)) - v_i(k, t)]], \quad k \in C \setminus \{i\},$$

where $s(t)$ is the state at time t , $x(t)$ is the action that agent k takes at time t , and $\eta^v(t)$ is a stepsize parameter.

Furthermore, each agent i updates its estimate for all agents $k \notin \mathcal{A}(t)$ that are not active at time t by setting

$$u_i(s(t), x_k(t)) = 0$$

and

$$v_i(k, t+1) = [v_i(k, t) + \eta^v(t) [-v_i(k, t)]], \quad k \in C \setminus \{i\}.$$

Note that each agent $i \in \mathcal{A}$ updates its estimates in this way whether they are active or not in the group at time t .

6.4.3 *Learning Observed Reputation.* Each agent i initializes its estimate of the observed reputation of other agents' behavior by setting

$$s_i(k, 0) = 0 \quad \text{for all } k \in C \setminus \{i\}.$$

All agents $i \in \mathcal{A}_p(t)$ that are active as participants at time t , interact and exchange their estimates of the observed reputation of other agents.

Recall that under the interaction rate $\mu_{p,i}(t)$, agent i is active as a participant in the group with probability $\theta(\mu_{p,i})$. In the following we define how the agents in $\mathcal{A}_p(t)$ interact in order to update their estimates of the agent with the highest observed reputation. To do that, we first define the information that they exchange, and then define how the two agents use this information to update their estimates of the observed reputation.

Estimate Update. Agents $i \in \mathcal{A}_p(t)$ update their reputation estimates $\{s_i(k, t)\}_{k \in \mathbb{B}(t) \setminus i}$ by taking the average estimates over all agents $j \in \mathcal{A}(t)$, and setting

$$s_i(k, t+1) = \frac{\sum_{j \in \mathbb{B}(t) \setminus k} s_j(k, t)}{|\mathcal{A}(t)|} + v_i(k, t+1) - v_i(k, t), \quad k \in \mathbb{B}(t) \setminus i. \quad (9)$$

If agent i is not active at time t , and $i \notin \mathcal{A}(t)$, then its estimates for all other agents $k \neq i$ remain unchanged, and we set

$$s_i(k, t+1) = s_i(k, t).$$

6.4.4 *Identifying the Highest-Reputation Agent.* All agents $i \in \mathcal{A}_p(t)$ that are active as a participant at time t update their estimate of the agent with the highest observed reputation as follows.

Intuitively, agent's i estimate of the agent with the highest observed observation at time $t+1$, is the agent with the highest reputation estimate $s_i(k, t+1)$. However, in order to handle noise in reputation estimates, agents choose the preferred agent employing a tie threshold $\Delta > 0$: if multiple agents' reputations fall within Δ of the highest estimate, the follower selects one uniformly at random (avoiding overfitting to transient fluctuations). This promotes robustness in dynamic environments. Formally, let $\Delta > 0$ be a tie threshold and let

$$\mathcal{K}_i(t+1) = \left\{ k \in C \setminus \{i\} : s_i(k, t+1) \geq \max_{k' \neq i} s_i(k', t+1) - \Delta \right\}.$$

Then the estimate of the agent with the highest observed reputation $L_i(t+1)$ is chosen uniformly at random from the set $\mathcal{K}_i(t+1)$.

If agent i is not active at time t , and $i \notin \mathcal{A}(t)$, then its estimates for all other agents $k \neq i$ remain unchanged, and we set

$$L_i(t+1) = L_i(t).$$

6.4.5 *Optimizing Reputation.* To optimize its reputation, agent i follows the agent k identified by $L_i(t+1)$ and adopts k 's behavior directly. Formally, this means

$$\phi(w_i^r(t+1); s, x) = \phi(w_k(t); s, x), \quad s \in \mathcal{S}, \quad x \in \mathcal{X}_s.$$

where $w_k(t)$ are the weights that agent k uses at time t to determine its behavior, i.e. which action x to choose in state $s(t)$. Concretely, if agent k optimizes its personal utility at time t and $k \in \mathbb{P}(t)$, then we have that

$$w_k(t) = w_k^{pu}(t),$$

otherwise we have that

$$w_k(t) = w_k^s(t),$$

This emulation allows agent i to acquire the reputation benefits associated with agent k 's norm.

6.5 Optimizing Status

Finally, we define how an agent $i \in \mathcal{A}_a(t)$ that is active as an actor in the group at time t updates its weights $w_i^s(t)$ to maximize its status.

6.5.1 Objective Function. In this case, at time t , agent k seeks to choose the parameter w_k^s that maximizes its expected social support from followers $\mathbb{F}_k(t)$ across states and actions:

$$J^s(w_k^s, t) = \sum_{s \in S} p(s) \sum_{x \in \mathcal{X}_s} \pi(w_k^s; s, x) \sum_{i \in \mathbb{F}_k(t)} \theta(\mu_i(t)) u_k(s, x). \quad (10)$$

6.5.2 Instantaneous Social Support. In the following, agent's use the social support obtained at time t to estimate the objective function $J^s(w_k^s, t)$. We refer to this support as the instantaneous social support received by agent i at time t . This instantaneous social support is a noisy observation of the actual social support given by $J(w_k^s, t)$.

Formally, when k chooses action $x(t)$ in state $s(t)$, the instantaneous social support that agent k receives at time t from a follower $i \in \mathbb{F}_k(t)$ is given by

$$D_{i,k}(s(t), x(t), t) = I_i(t) u_i(s(t), x(t)), \quad (11)$$

where $I_i(t)$ is the indicator whether agent i is active in the group at time t as an participant, i.e., whether $i \in \mathcal{A}_p(t)$.

The total instantaneous social support received for the action action $x(t)$ in state $s(t)$ at time t is then given by

$$\sum_{i \in \mathbb{F}_k(t)} D_{i,k}(s(t), x(t), t),$$

6.5.3 Gradient Estimation and Update. Using the total instantaneous support, agent k then uses again a standard stochastic gradient algorithm to maximize the objective function $J(w_k, t)$. That is, given the state $s(t)$ at time t , agent k first chooses an action $x_k(t)$ where the action $x_k(t)$ randomly according to the random policy $\pi(w_k^s(t); s(t), x)$ for the current weight vector $w_k^s(t)$. Next, agent i updates its weight vector by setting

$$w_k^s(t+1) = w_k^s(t) + \beta(t) \cdot \nabla_w \log \pi(w_k^s(t); s(t), x_k(t)) \cdot \sum_{i \in \mathbb{F}_k(t)} D_{i,k}(s(t), x_k(t), t), \quad (12)$$

where $\beta(t)$ is a stepsize parameter.

This learning rule increases the likelihood of actions that elicit stronger positive responses (i.e., higher utility) from followers, thereby encouraging norms that are broadly beneficial and socially rewarded.

6.6 Updating Reward Estimates

Let $\hat{J}_i^{pu}(t)$, $\hat{J}_i^r(t)$, and $\hat{J}_i^s(t)$, be the estimates of the rewards obtained in the different roles. Agent i initializes these estimates at time $t = 0$ by setting them equal to 0, and then uses the learning algorithm provided in this section to update these estimates at time $t \geq 1$.

Moreover, agent i will use these estimates both to decide which role to take on, as well as decide on its interaction rates $\mu_{a,i}(t)$ and $\lambda_{a,i}(t)$.

Updating Estimate of Personal Utility. All agents such that $i \in \mathbb{P}(t)$ and $i \in \mathcal{A}_a(t)$ that are active as an actor at time t and optimize their personal utility at time t use the observed personal benefit

$$u_i(s(t), x_i(t))$$

to update their estimate of the reward for personal utility by setting

$$\hat{J}_i^{pu}(t) = \hat{J}_i^{pu}(t-1) + \eta^J(t) [u_i(s(t), x_i(t)) - \hat{J}_i^{pu}(t-1)]$$

where $\eta^J(t)$ is a stepsize parameter.

Updating Estimate of Reward from Reputation. All agents $i \in \mathbb{R}(t)$ and $i \in \mathcal{A}_a(t)$ that are active as an actor at time t and that optimize their reputation at time t update their estimate as

$$\hat{J}_i^r(t) = s_i(k, t),$$

where k is the agent that i follows in order to optimize its reputation.

Updating Estimate of Reward from Status. All agents $k \in \mathcal{A}_a(t)$ that are active as an actor at time t and have followers at time t ($\mathbb{F}_k(t) \neq \emptyset$) use the instantaneous social support

$$\sum_{j \in \mathbb{F}_k(t)} D_{j,k}(s(t), x(t), t)$$

to update their estimate of the reward for status by setting

$$\hat{J}_k^s(t) = \hat{J}_k^s(t-1) + \eta^J(t) \left[\sum_{j \in \mathbb{F}_k(t)} D_{j,k}(s(t), x(t), t) - \hat{J}_k^s(t-1) \right],$$

where $\eta^J(t)$ is a stepsize parameter.

Unchanged Estimates. All other estimates that are not updated according to the above rules remain unchanged; that is, their value at time t equals their value at time $t-1$.

Discussion. Note that while most agents update only a single reward estimate at each time t , there is one important exception: agents that optimize their personal utility and have followers ($\mathbb{P}(t) \ni i$ and $\mathbb{F}_i(t) \neq \emptyset$) update both their estimate for personal utility and their estimate for status. This reflects the fact that such agents not only observe their own direct utility but also receive social feedback from followers, allowing them to refine both estimates simultaneously.

6.7 Optimizing Interaction Rates as an Actor

Let

$$\hat{H}_i(t) = \max\{\hat{J}_i^{pu}(t), \gamma \hat{J}_i^r(t), \kappa \hat{J}_i^s(t)\}.$$

We set

$$\mu_{a,i}(t) = [\mu_{a,i}(t-1) + \alpha(t) [-\theta' (M - \mu_{a,i}(t-1)) u_0 + \theta' (\mu_{a,i}(t-1)) \hat{H}_i(t)]]_0^M, \quad (13)$$

where $u_0 > 0$ is a given constant that reflects the utility that an agent obtains from participating as an actor outside the group, and

$$[x]_0^M = \min\{\max(x, 0), M\}, \quad x \in \mathbb{R}.$$

6.8 Discussion

The algorithm we propose presents several challenges. First, it introduces a novel reputation learning process that runs in parallel with the agents' reward optimization algorithms. Second, agents can switch roles, and each role corresponds to a different reward component. As a result, the algorithm does not optimize a continuous objective function: each role switch introduces a "jump" or discontinuity in the objective.

To address these challenges and keep the analysis and notation tractable — which are already non-trivial even with simplifying assumptions — we adopt a few simplifications. Importantly, these

assumptions can be relaxed without affecting our main results. In the following, we describe these assumptions and discuss how they could be lifted.

6.8.1 Reputation Learning. To simplify both the notation and the analysis, we assume in this paper that all agents $i \in \mathcal{A}(t)$ that are active at time t participate in the reputation learning algorithm defined in Subsection 6.3.2. However, this assumption can be relaxed in two important ways:

- (1) **Random subset interactions:** Instead of assuming that all active agents interact at each time step, we can consider the case where only a random subset of agents $i \in \mathcal{A}(t)$ engage in interactions to exchange their estimates of which agent has the highest observed reputation.
- (2) **Connectivity graph:** We can assume that agents interact according to an underlying connectivity graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where the nodes $\mathcal{V}$ correspond to the set of agents $\mathcal{A}$, and $\mathcal{E}$ defines the edges between agents. At time t , two agents i and j exchange their estimates only if both are active ($i, j \in \mathcal{A}(t)$) and there exists an edge $(i, j) \in \mathcal{E}$ in the graph $\mathcal{G}$.

Our main results continue to hold under either of these relaxed assumptions, provided that:

- In the first case, each active agent $i \in \mathcal{A}(t)$ has a probability of at least $p_0 > 0$ of being selected for interaction at each time step.
- In the second case, the connectivity graph $\mathcal{G}$ is connected (i.e., there exists a path between any pair of agents).

Both of these conditions are mild and align with standard assumptions in the distributed learning literature. They ensure that information about the highest reputation agent eventually propagates throughout the population. Importantly, relaxing the assumption that all active agents participate at each step does not affect the convergence guarantees of the reputation learning algorithm or the main results of our analysis.

6.8.2 Reputation Optimization. The approach to reputation optimization presented in this work intentionally adopts the simplest possible mechanism: once an agent i identifies another agent k as having the highest reputation, it directly copies k 's policy parameters. This hard imitation ensures that agent i inherits the same norm as k , and thereby gains similar reputational benefits.

However, more sophisticated reputation learning schemes could be employed. For example, instead of directly copying the policy parameters of agent k , agent i could observe the actions taken by k across a sequence of states and train its own policy representation to mimic this behavior. In this approach, agent i would maintain a dataset of state-action pairs (s, x_k) based on observations of agent k , and update its own parameters via supervised learning to minimize a divergence between its policy and the observed behavior:

$$\min_{w_i} \sum_{(s, x_k)} -\log \pi(w_i; s, x_k).$$

This imitation learning framework, often referred to as behavioral cloning, enables smoother adaptation and allows for partial or noisy observations. While we do not implement this in our current model, such techniques could enhance robustness and learning efficiency in more complex or partially observable environments.

Importantly, such a learning-based imitation scheme can be easily incorporated into our algorithm and analysis using standard techniques from online learning and policy optimization. However, to keep the exposition and notation streamlined—and to avoid additional overhead in the theoretical analysis—we adopt the simpler direct-copying rule in this work. This choice allows us to highlight the essential structure and convergence properties of the overall system without distraction from secondary implementation details.

7 UPDATING ROLES

In this section we define how agents decide which role to take on. This process mirrors the role decision rules given in Section ??, but here agents use estimates $\hat{j}_i^{pu}(t)$, $\hat{j}_i^r(t)$, and $\hat{j}_i^s(t)$ of the payoffs under different reward components, rather than the actual payoffs.

7.1 Definitions and Notation

In this subsection we introduce additional definitions and notation that we use in this section.

7.1.1 Roles and Followers. Recall that $\mathbb{P}(t)$, $\mathbb{R}(t)$, and $\mathbb{S}(t)$, indicate which agents optimize their personal utility, reputation, and status, respectively, at time t . At each time step t , each agent decides on a single reward component to optimize, and we have that these sets are disjoint with

$$\mathbb{P}(t) \cup \mathbb{R}(t) \cup \mathbb{S}(t) = \mathcal{A}, \quad t \geq 0.$$

All agents start out by optimizing their personal utility, and we have that

$$\mathbb{P}(0) = \mathcal{A},$$

and $\mathbb{R}(0) = \mathbb{S}(0) = \emptyset$.

Also recall that $\mathbb{F}_k(t)$ denotes the set of agents $i \in \mathbb{R}(t)$ that follow agent k at time t to optimize their reputation. Initially we have that

$$\mathbb{F}_k(t) = \emptyset, \quad k \in \mathcal{A}.$$

In the following, we define how the sets $\mathbb{P}(t)$, $\mathbb{R}(t)$, and $\mathbb{S}(t)$, and the sets $\mathbb{F}_k(t)$, $k \in \mathcal{A}$, evolve over time as agents decide on which role to take on.

7.1.2 Threshold on Number of Followers. An agent considers optimizing its status only if it has sufficiently many followers:

$$|\mathbb{F}_i(t)| \geq cN,$$

where $c \in [0, 1]$ is a given constant.

This ensures that status decisions are based on reliable social feedback. With too few followers, feedback may misrepresent community preferences, leading to actions that reduce status. Requiring a larger base of followers increases the chance that behavior changes genuinely enhance both social welfare and the agent's status.

7.1.3 Threshold on Reputation. We introduce two reputation thresholds, B_R and B_F , satisfying

$$0 < B_F < B_R,$$

that agents use when deciding whether to follow another agent. Specifically:

- a) An agent will decide to follow another agent only if that agent's reputation is at least B_R .
- b) Once an agent is a follower, it will continue following as long as the reputation of the agent with the highest reputation remains above B_F .

This assumption creates a hysteresis: it takes a sufficiently high reputation to start following, but a larger drop before an agent will stop following. This prevents agents from switching back and forth when the highest reputation oscillates around B_R .

7.1.4 Update Epochs. Agents periodically re-evaluate which role to adopt. Let T_n be a sequence of positive integers indicating the interval between updates. For simplicity, we assume all agents use the same sequence $\{T_n\}_{n \geq 1}$, though this can be relaxed (see Subsection ??).

Setting $s_0 = 0$, the n th update epoch is

$$s_n = s_{n-1} + T_n, \quad n \geq 1.$$

At epoch s_n , each agent i decides which reward component to optimize during the interval $t = s_n + 1, \dots, s_{n+1}$ based on its payoff estimates.

7.2 Sequential Role Update Procedure

At each epoch s_n , all agents simultaneously decide which role to adopt. Since these decisions are interdependent—especially for reputation optimization—the updates must be applied in a carefully chosen order to avoid inconsistencies.

For example, if agent i considers following agent j , but j is already following agent k , then i should ultimately follow k rather than j . Proper ordering ensures that such indirect relationships are respected: each agent observes the most up-to-date follower sets when making its decision.

The procedure updates roles sequentially, with each step using the role sets $\mathbb{P}$, $\mathbb{R}$, $\mathbb{S}$, and follower sets $\mathbb{F}_k$ produced by all previous steps. Formally:

- (1) *Step 0 – Initialization:* Copy the current role and follower sets:

$$\mathbb{P}(s_n) = \mathbb{P}(s_{n-1}), \quad \mathbb{R}(s_n) = \mathbb{R}(s_{n-1}), \quad \mathbb{S}(s_n) = \mathbb{S}(s_{n-1}), \quad \mathbb{F}_k(s_n) = \mathbb{F}_k(s_{n-1}), \quad \forall k \in \mathcal{A}.$$

- (2) *Step 1 – Reputation:* Agents without followers decide whether to optimize reputation starting at $s_n + 1$. This updates $\mathbb{P}(s_n)$, $\mathbb{R}(s_n)$, and $\mathbb{F}_k(s_n)$.

- (3) *Step 2 – Status:* Agents with sufficiently many followers and favorable status estimates update their roles, modifying $\mathbb{S}(s_n)$ and $\mathbb{P}(s_n)$.

- (4) *Step 3 – Personal Utility:* All remaining agents default to personal-utility optimization. This finalizes $\mathbb{P}(s_n)$, $\mathbb{R}(s_n)$, $\mathbb{S}(s_n)$, and $\mathbb{F}_k(s_n)$.

This sequential, in-place procedure ensures consistency and eliminates ambiguities from indirect follower relationships. We now define each of the three steps in detail.

7.3 Step 1 - Optimizing Reputation

At time $t = s_n$, agents without followers evaluate whether their estimated reward from optimizing reputation, $\gamma \hat{J}_i^r(s_n)$, exceeds the reward from personal utility, $\hat{J}_i^{pu}(s_n)$. If so, the agent follows the one it believes currently has the highest reputation. To maintain consistent follower relationships, agents never follow someone who is already a follower. The decision also uses one of two reputation thresholds: B_R for agents not yet following anyone, and B_F for agents who are already followers, creating a hysteresis effect that stabilizes following behavior as discussed above. The precise update procedure is as follows.

Let $\mathbb{C}(s_n)$ be the set of agents that do not have a follower at time $t = s_n$. We further partition it into $\mathbb{C}^r(s_n)$, the agents already following another agent, and $\mathbb{C}^{pu}(s_n)$, the agents not yet following anyone.

For each agent $i \in \mathbb{C}(s_n)$, define the effective threshold

$$B_i = \begin{cases} B_F, & i \in \mathbb{C}^r(s_n) \\ B_R, & i \in \mathbb{C}^{pu}(s_n) \end{cases}$$

to decide whether to follow or remain following.

Let $N_c = |\mathbb{C}(s_n)|$, and let $\sigma = (\sigma(1), \dots, \sigma(N_c))$ be a permutation representing the update order. For $q = 1, \dots, N_c$, let $i = \sigma(q)$ and perform the following updates:

- (1) If

$$\gamma \hat{J}_i^r(s_n) > \max\{B_i, \hat{J}_i^{pu}(s_n)\},$$

then agent i adopts reputation optimization for $t = s_n + 1, \dots, s_{n+1}$ by following

$$\hat{k} = L_i(s_n),$$

the agent that i believes has the highest observed reputation.

(2) If $\hat{k} \in \mathbb{F}_{k'}(s_n)$ for some k' , then agent i follows k' and we set $k = k'$. Otherwise, $k = \hat{k}$.

(3) Update the role sets:

$$\mathbb{R}(s_n) = \mathbb{R}(s_n) \cup \{i\}, \quad \mathbb{P}(s_n) = \mathbb{P}(s_n) \setminus \{i\}.$$

(4) Update the follower sets:

$$\mathbb{F}_k(s_n) = \mathbb{F}_k(s_n) \cup \{i\}, \quad \mathbb{F}_j(s_n) = \mathbb{F}_j(s_n) \setminus \{i\}, \quad \forall j \neq k.$$

This procedure ensures that all agents without followers are updated sequentially and never follow an agent who is already a follower, preserving the consistency of indirect follower relationships.

7.4 Step 2 - Optimizing Status

At time $t = s_n$, if agent i has a sufficiently large number of followers,

$$|\mathbb{F}_i(s_n)| \geq cN,$$

and its estimated reward from optimizing status exceeds that from personal utility, i.e.,

$$\kappa \hat{j}_i^s(s_n) > \hat{j}_i^{pu}(s_n),$$

then agent i will optimize its status during $t = s_n + 1, \dots, s_{n+1}$. The role sets are updated as:

$$\mathbb{S}(s_n) = \mathbb{S}(s_n) \cup \{i\}, \quad \mathbb{P}(s_n) = \mathbb{P}(s_n) \setminus \{i\}.$$

7.5 Step 3 - Optimizing Personal Utility

If agent i does not meet the conditions to optimize reputation or status, it will optimize its personal utility during $t = s_n + 1, \dots, s_{n+1}$. We update:

$$\mathbb{P}(s_n) = \mathbb{P}(s_n) \cup \{i\},$$

$$\mathbb{R}(s_n) = \mathbb{R}(s_n) \setminus \{i\}, \quad \mathbb{S}(s_n) = \mathbb{S}(s_n) \setminus \{i\}.$$

For all agents $j \neq i$ we set

$$\mathbb{F}_j(s_n) = \mathbb{F}_j(s_n) \setminus \{i\}.$$

7.6 Discussion

The threshold parameter c , which governs the decision to switch from optimizing personal utility to optimizing status, plays a crucial role. If c is set too low, agents may switch prematurely to optimizing status, leading to reduced reputation and a potential loss of followers. If c is too high, agents may delay switching even when status optimization would be beneficial.

The appropriate value of c depends on factors such as group size N and the personal utility functions $u_i(s, x)$ that agents obtain from different states and actions. As such, the optimal value of c cannot be known in advance and must be learned from social feedback. Specifically, if an agent switches to optimizing status and subsequently experiences a decline in its number of followers, this indicates that the agent's contribution may no longer align with the social reward structure. In such cases, the agent should revert to optimizing personal utility and increase the threshold c , making it less likely to switch to status optimization prematurely in the future. Conversely, if the agent's follower count remains stable or increases, the current value of c can be maintained or even slightly decreased. This adaptive process allows agents to calibrate role transitions dynamically in response to their social environment, fostering a balance between self-interest and socially beneficial behavior.

Although incorporating learning dynamics for the threshold parameter c is possible, we omit this analysis to avoid overloading the algorithm definition and theoretical analysis, which are already complex. We leave this extension for future work.

8 STEPSIZE PARAMETERS, UPDATE INTERVALS, AND TIME-SCALES

The convergence and stability of the learning dynamics in our model critically depend on the appropriate choice of stepsize parameters and the careful design of a time-scale hierarchy across the different update processes. Because agents simultaneously optimize multiple objectives—including personal utility, reputation, and status—while also adapting their interaction rates and roles, ensuring that these learning processes evolve on appropriately separated time-scales is essential for preventing destabilizing feedback loops and guaranteeing convergence to a stable equilibrium.

In this section, we formally introduce the stepsize parameters associated with each learning process and specify the assumptions required for their convergence. We begin by defining the time-scale hierarchy, outlining the ordering of the different processes based on their update frequencies. We then state the assumptions on the stepsize sequences and update intervals that enforce these separations in time-scales. Together, these conditions ensure that faster processes, such as reputation learning, stabilize before slower processes, such as behavior optimization and role adaptation, make significant updates—thereby enabling convergence of the overall coupled learning system.

8.1 Two Time-Scale Approach

The learning dynamics in our model unfold across two main categories of processes given as follows:

- (1) *Continuous Processes*: These are processes in which agents make updates whenever they are active in the group. They include the gossip process, the process for updating the agents' behavior, and the process for updating the interaction rates.
- (2) *Discrete Processes*: These are processes where updates occur only periodically, rather than every time an agent is active. This includes the processes for updating the agents' roles.

To ensure convergence to a learning outcome, these processes need to evolve on different time-scales: some processes need to evolve on a fast time-scale, allowing agents to quickly adapt to new feedback, while other processes need to evolve on a slow time-scale, enabling more stable structural adaptation and role differentiation. The positions of the different processes with respect to these two time-scales are given as follows:

- (1) *Reputation Learning and Reward Estimation (Continuous Process on Fast Time-Scale)*: Reputation learning as defined in Section ?? and reward estimation as defined in Section ??, occur on the fast time-scale. This includes estimating the personal benefit that agent i receives from another agent k , and identifying the agent with the highest observed reputation, and updating the reward estimates for the different roles. The assumption that these processes evolve on the fast time-scale is formalized in Assumption ?? and Assumption ??.
- (2) *Behavior Optimization (Continuous Process on Slow Time-Scale)*: Behavior optimization, described in Sections ?? and ??, occurs on a slow time-scale. This ensures that updates of the weights using gradient estimates tracks an ODE. The separation of this time-scale is ensured by Assumption ??.
- (3) *Interaction Rate Optimization (Continuous Process on Slow Time-Scale)*: Interaction rate optimization as defined in Section ??, occur on a slow time-scale. This ensures that the rates do not change drastically between role updates. The separation of this time-scale is ensured by Assumption ??.
- (4) *Role Optimization (Discrete Process on Slow Time-Scale)*: Finally, role optimization as defined in Section ?? occurs on a slow time-scale. This process involves determining whether to optimize personal utility, reputation, or status, and they only change after an accurate estimate of the rewards that are obtained for each role is accumulated. The sufficiently slow evolution of these processes is enforced by Assumptions ?? and ??.

This separation of time-scales ensures that lower-level dynamics stabilize before higher-level decisions are made, thereby enabling convergence in the presence of coupled learning processes.

8.2 Assumptions on Stepsize Parameters

Recall that $\alpha(t)$ is the setsize parameter used to update the weights $w_i^{pu}(t)$, $i \in \mathcal{A}$, for the behavior used to optimize the personal utility, $\beta(t)$ is the setsize parameter used to update the weights $w_i^s(t)$, $i \in \mathcal{A}$, for the behavior used to optimize the status, $\eta^v(t)$ is the setsize parameter used to update the estimates $v_i(k, t)$ of the benefit that the behavior of agent k provides agent i , and $\eta^J(t)$ is the setsize parameter that agents use to update their estimate of the personal utility $\hat{J}_i^{pu}(t)$ and the rewards from status $\hat{J}_i^{pu}(t)$.

We then make the following standard assumption for these stepsize parameters.

ASSUMPTION 5. *The following properties hold for the stepsize parameters.*

(1) *The stepsizes $\alpha(t)$, $\alpha(t) > 0$, satisfy the conditions*

$$\sum_{t=1}^{\infty} \alpha(t) = \infty, \quad \text{and} \quad \sum_{t=1}^{\infty} \alpha(t)^2 < \infty.$$

(2) *The stepsizes $\beta(t)$, $\beta(t) > 0$, satisfy the conditions*

$$\sum_{t=1}^{\infty} \beta(t) = \infty, \quad \text{and} \quad \sum_{t=1}^{\infty} \beta(t)^2 < \infty.$$

(3) *The stepsizes $\eta^v(t)$, $\eta^v(t) > 0$, satisfy the conditions*

$$\sum_{t=1}^{\infty} \eta^v(t) = \infty, \quad \text{and} \quad \sum_{t=1}^{\infty} \eta^v(t)^2 < \infty.$$

(4) *The stepsizes $\eta^J(t)$, $\eta^J(t) > 0$, satisfy the conditions*

$$\sum_{t=1}^{\infty} \eta^J(t) = \infty, \quad \text{and} \quad \sum_{t=1}^{\infty} \eta^J(t)^2 < \infty.$$

8.3 Assumptions on Update Intervals

We make the following assumption for the update intervals T_n and T_n^{rate} .

ASSUMPTION 6. *For the update intervals $\{T_n\}_{n \geq 1}$ we have that*

$$\lim_{n \rightarrow \infty} T_n = \infty, .$$

8.4 Assumptions on Time-Scales

The next assumption ensures that the behavior learning using the stepsiz parameters $\alpha(t)$ and $\alpha(t)$ evolves on a slower time-scale than estimating the personal benefit and the obtained rewards under the different roles that uses the stepsize parameters $\eta^v(t)$ and $\eta^J(t)$.

ASSUMPTION 7. *For the stepsizes $\alpha(t)$, $\beta(t)$, $\eta^v(t)$, and $\eta^J(t)$, we have that*

(1)

$$\lim_{t \rightarrow \infty} \alpha(t)/\eta^v(t) = 0$$

(2)

$$\lim_{t \rightarrow \infty} \alpha(t)/\eta^J(t) = 0$$

(3)

$$\lim_{t \rightarrow \infty} \beta(t)/\eta^v(t) = 0$$

(4)

$$\lim_{t \rightarrow \infty} \beta(t)/\eta^J(t) = 0$$

Our final assumption characterizes how the time-scale of update intervals T_n^{rate} compare with the time-scale of the stepsizes $\alpha(t)$, $\beta(t)$, $\eta^v(t)$, and $\eta^J(t)$,

ASSUMPTION 8. *For the stepsizes $\alpha(t)$, $\beta(t)$, $\eta^v(t)$, and $\eta^J(t)$, we have for all agents $i \in C$ that*

(1)

$$\lim_{n \rightarrow \infty} \sum_{t=s(n)+1}^{s(n+1)} \eta^v(t) = \infty.$$

(2)

$$\lim_{n \rightarrow \infty} \sum_{t=s(n)+1}^{s(n+1)} \eta^J(t) = \infty.$$

(3)

$$\lim_{n \rightarrow \infty} \sum_{t=s(n)+1}^{s(n+1)} \alpha(t) = 0.$$

(4)

$$\lim_{n \rightarrow \infty} \sum_{t=s(n)+1}^{s(n+1)} \beta(t) = 0.$$

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